

CLASSIFYING IRISH TUNE TYPES FROM ABC NOTATION WITH A CHARACTER-LEVEL LSTM

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ABSTRACT

Irish traditional tune types (reels, jigs, hornpipes, etc.) have been classified using hand-crafted symbolic features or audio embeddings of synthesized scores; neural sequence models trained on the same ABC-notation corpora have so far been used only for tune generation, not classification. This work trains a character-level LSTM directly on raw ABC-notation transcriptions from The Session to classify tune type across twelve categories, and compares it against a support-vector-machine baseline built on music21-derived pitch, interval, and rhythm histograms, following prior symbolic-feature classifiers for this task. Both models are evaluated on a piece-level train/validation/test split to avoid leakage between settings of the same tune. Without class-weighted training, the LSTM reaches 83.9% test accuracy, exceeding the SVM baseline’s 82.7%, but the SVM retains a higher macro-F1 (0.739 vs. 0.692), reflecting a trade-off between overall accuracy and balanced performance on metrically similar minority classes. A lightweight neural sequence model can match a hand-engineered baseline directly from text, without audio synthesis or manual feature design, while class imbalance remains an open challenge.

1. INTRODUCTION

Irish traditional tune types are distinguished primarily by meter and rhythmic feel rather than pitch content, and correctly labelling a tune with its type is a routine task for archives and community repositories such as The Session [1], which hosts tens of thousands of tunes in ABC notation, a compact text-based symbolic format. Prior work has approached this task with classical classifiers on hand-crafted symbolic features: Martins and Silla trained decision trees reaching 92.3% overall precision across eleven genres, with substantially lower precision on metrically similar types such as barndance [2], while Kermit-Canfield and Roman found that reels and hornpipes, which share a similar meter, remained close to chance even for the best classifier [3]. Other work has used deep embeddings of

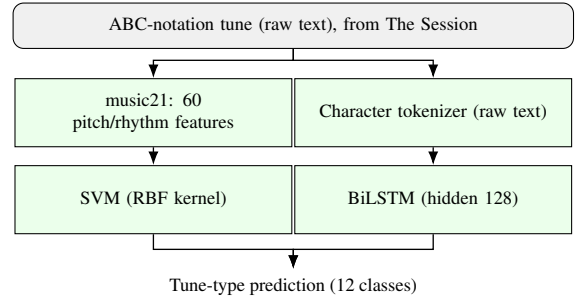


Figure 1. The two classification pipelines compared in this work: a feature-engineered SVM baseline and an end-to-end LSTM trained directly on raw ABC text.

audio synthesized from scores [4], and, most recently, data augmentation applied directly to symbolic ABC scores [5]. Separately, neural sequence models have been trained on the same ABC corpora, but only for tune generation [6, 7]. To the best of the author’s knowledge, no prior work trains a neural sequence model end-to-end and directly on raw ABC text for tune-type classification, without synthesizing audio or engineering features by hand. This work addresses that gap and reports where a character-level LSTM succeeds and fails relative to a feature-based SVM baseline. Code and additional results are available online ¹.

2. METHOD

Figure 1 summarizes the two pipelines compared in this work. Because reels, jigs, and hornpipes are set apart mainly by meter and rhythmic feel rather than melodic content, a classifier for this task must learn rhythmic structure directly from the transcription.

Data. Tunes and their type labels are drawn from The Session [1] (`tunes.csv`), yielding 55,352 settings across 23,300 unique tunes and 12 tune types after parsing, with a substantial class imbalance (reel: 20,187; three-two: 428). Data are split 70/15/15 into train/validation/test at the tune level, not the setting level, so that no tune appears in more than one split, following the leakage-avoidance practice established for symbolic music classification [8].

Baseline. Following [2, 3], a support-vector machine is trained on 60 hand-crafted features per tune extracted with music21 [9]: pitch-class and melodic-interval histograms,

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¹ <https://mariovallejotec.github.io/irish-tune-classification/>

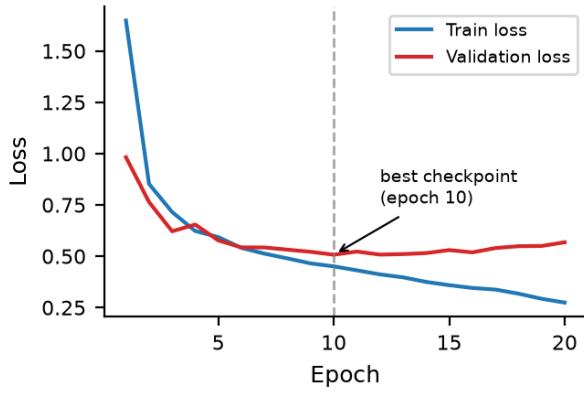


Figure 2. LSTM train/validation loss by epoch. Validation loss reaches its minimum at epoch 10, after which training loss keeps decreasing while validation loss rises.

melodic direction statistics, note-duration histograms, and meter.

Neural model. A single-layer bidirectional LSTM (hidden size 128), implemented in PyTorch, is trained directly on the raw ABC character sequence (no feature extraction, max length 600), with a linear classification head. The full model totals 209,804 parameters. Training uses AdamW [10] with warmup and cosine decay, gradient clipping, dropout (0.3) and weight decay (10^{-4}), and early stopping on validation loss (patience 10; Fig. 2) – hyperparameter choices informed by common practice across recent neural network papers [11–13], since no prior work addresses class imbalance for a neural sequence model on this exact task.

3. RESULTS

Model	Accuracy	Macro-F1
SVM (RBF)	82.7%	0.739
LSTM (class-weighted)	78.4%	0.653
LSTM (unweighted)	83.9%	0.692

Table 1. Test-set results for the SVM baseline and the LSTM under class-weighted and unweighted training.

The unweighted LSTM’s higher accuracy comes from strong performance on majority classes (reel F1 0.882, jig F1 0.966), while precision and recall collapse on tune types that share a meter with more frequent classes (barndance: precision 0.383, recall 0.303; march: precision 0.505, recall 0.302). This is not an arbitrary error pattern (Fig. 3): reel, hornpipe, and barndance are all played in 4/4 and share a similar eighth-note pulse, whereas jig’s distinct 6/8 compound feel gives the model an unambiguous rhythmic cue – the same reel/hornpipe-type confusions reported for the SVM baseline in [3]. Class-weighted training raises minority-class recall but lowers overall accuracy, indicat-

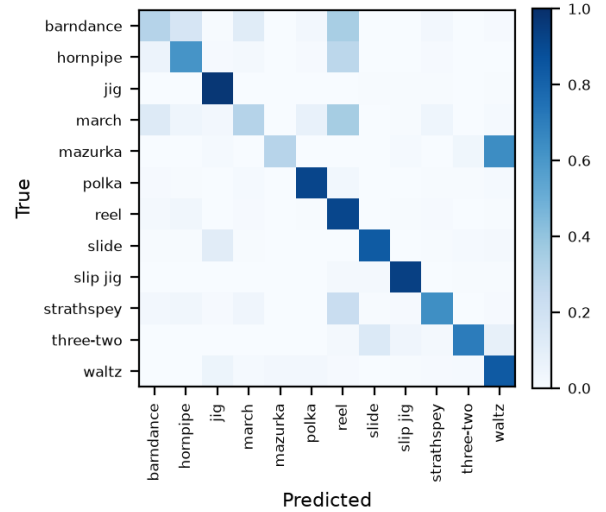


Figure 3. Confusion matrix of the unweighted LSTM on the test set (row-normalized).

ing that class imbalance, not architecture choice, is the main remaining obstacle for this task.

Ablation. Table 2 evaluates the contribution of class-weighted loss, regularization (dropout and weight decay), and symbolic transposition augmentation [5] to the LSTM’s performance. Augmentation lowers both accuracy and macro-F1 relative to the unaugmented model. Regularization alone leaves test performance almost unchanged while class weighting is on; only removing class weighting, on top of regularization, produces a substantial shift toward higher accuracy at the cost of macro-F1.

Class w.	Reg.	Aug.	Accuracy	Macro-F1
✓	✗	✗	77.5%	0.650
✓	✗	✓	76.5%	0.628
✓	✓	✗	78.4%	0.653
✗	✓	✗	83.9%	0.692

Table 2. Ablation over the LSTM training recipe: class-weighted loss (Class w.), regularization (Reg.: dropout + weight decay), and transposition augmentation (Aug.).

4. CONCLUSION

A character-level LSTM trained directly on raw ABC-notation text can match or exceed a feature-engineered SVM baseline in accuracy on Irish tune-type classification, without audio synthesis or hand-crafted features, though it does not resolve the class-imbalance problem that the same baseline family already exhibited. Future work includes cross-validation, comparison against recent symbolic-augmentation approaches [5], and dedicated class-imbalance techniques for sequence models.

5. REFERENCES

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